Notes on Clustering (Kmeans):

— Clustring : The grouping of objects such that objects in the same cluster are more similar to each other than they are to objects in another cluster.

— The task of finding an assignment of data points to clusters, as well as a set of vectors {μk}, such that the sum of the squares of the distances of each data point to its closest vector μk (or another similarity measure) is a minimum.

– K in Kmeans relates to how many clusters you want to split your data into.

— What is the K-Means algorithm?

- Choose the number of clusters k
- Select k random points from the data as centroids
- Until Convergence

a. Assignment to a cluster Head/Centroid.

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$

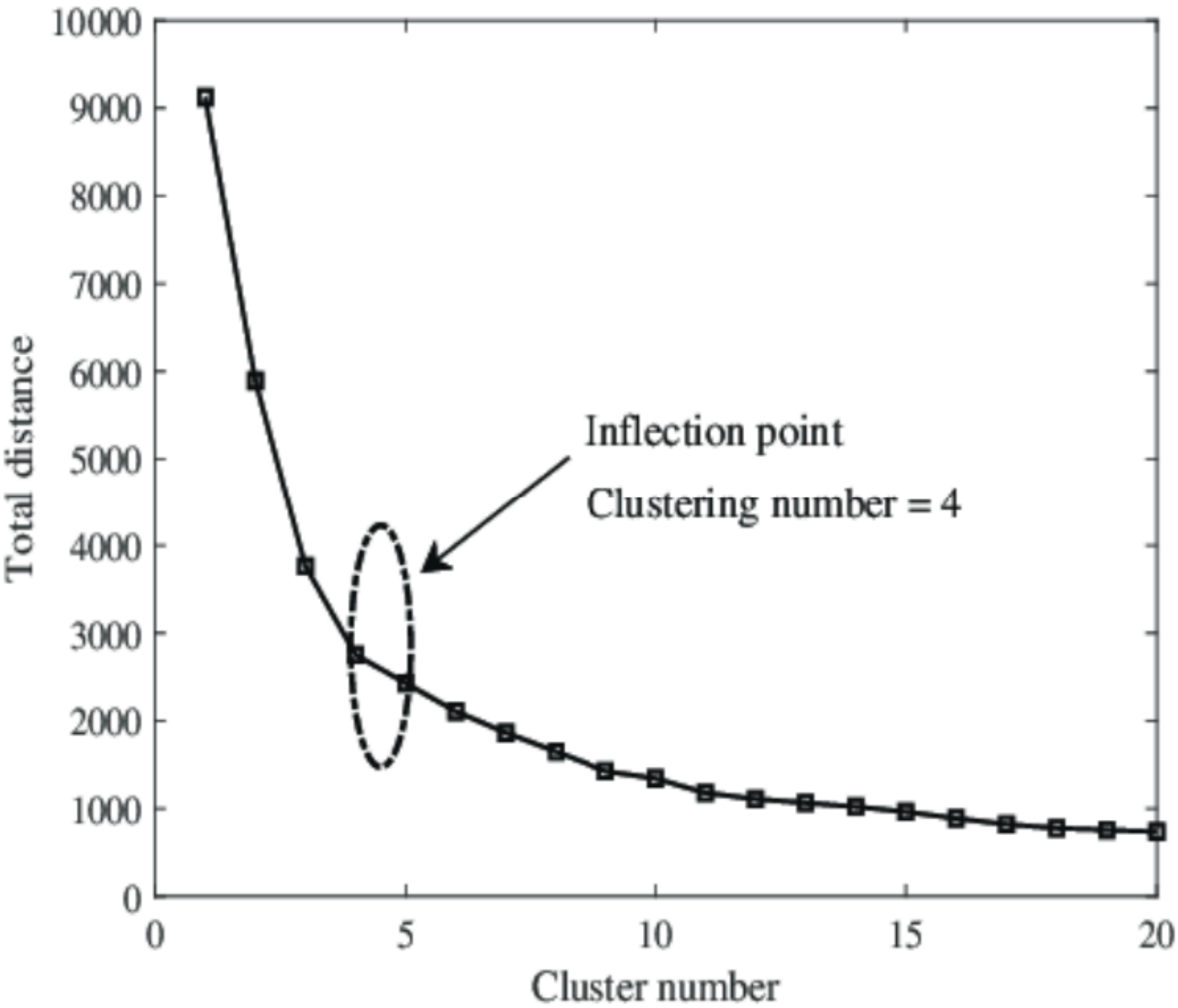
b. Resetting the Centroid.
Find the mean of all points in a cluster, and set that as the new centroid of the cluster. (Hence the name k-means)

$$\boldsymbol{\mu}_k = \frac{\sum_n r_{nk}\mathbf{x}_n}{\sum_n r_{nk}}.$$

– What is Convergence in K-Means? There is no change in the assignments of the centroid of clusters, or cluster assignment of points in the algorithm.

– How can we choose a number of Topics for K-Means?

1. The Most Common Approach is to Visualise it and Choose.
2. Using The Elbow Method.



– As the number of clusters (K) increases the average distance cluster distance decreases.

– Best choice for K around the elbow of the curve.

– Feature Scaling is required when you work with Kmeans.

– If you do not scale features, the feature with much larger variance will dominate the importance in clustering.